

PH ARVIND SHARMA

Data Scientist | Machine Learning Engineer

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PROFESSIONAL SUMMARY

Data Scientist with 2+ years of hands-on experience designing and deploying end-to-end machine learning pipelines across computer vision, NLP, time-series forecasting, and deep learning domains. Demonstrated success achieving 90–95% model accuracy on real-world datasets in healthcare, retail, and energy sectors. Proficient in Python, TensorFlow, PyTorch, Scikit-learn, and cloud-based deployment using Streamlit and Flask. Skilled in the complete ML lifecycle: data preprocessing, exploratory data analysis (EDA), feature engineering, model tuning, evaluation, and production deployment. Passionate about building scalable, high-impact AI solutions.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Machine Learning: Scikit-learn, XGBoost, Supervised Learning, Unsupervised Learning, Classification, Regression

Deep Learning: TensorFlow, PyTorch, CNN, LSTM, Transfer Learning, YOLOv8

NLP and LLMs: Natural Language Processing (NLP), Large Language Models (LLMs), LLaMA, Groq API, Hugging Face Transformers, Text Classification, Emotion Analysis, AI Agents

Time Series Forecasting: ARIMA, SARIMA, LSTM-based Forecasting, Demand Forecasting

Data Analysis and Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, EDA, Feature Engineering

Deployment and Tools: Streamlit, Flask, OpenCV, PostgreSQL, Microsoft Azure, GitHub, PyCharm

WORK EXPERIENCE

Data Science Intern (AI/ML) *InLighnX Global Pvt. Ltd.*

Oct 2025 – Apr 2026

- Engineered a Traffic Sign Recognition system using Convolutional Neural Networks (CNN) on the GTSRB dataset (50,000+ images, 43 classes), achieving 95% test accuracy through data augmentation and hyperparameter tuning.
- Built Walmart Sales Forecasting models using ARIMA, SARIMA, and LSTM, achieving 90–93% prediction accuracy; optimized seasonality and trend decomposition to reduce mean absolute percentage error (MAPE) by 8%.
- Designed and deployed an Energy Consumption Forecasting system combining LSTM and SARIMA models, achieving 92–95% prediction accuracy; integrated results into a real-time Streamlit dashboard for automated monitoring.
- Trained a Pneumonia Detection model using CNN on labeled chest X-ray datasets, achieving 92–95% diagnostic accuracy; applied class-imbalance handling and transfer learning to improve sensitivity by 12%.
- Developed ANAKIN, a multimodal real-time AI assistant integrating Speech-to-Text (STT), Text-to-Speech (TTS), OpenCV, and LLMs via Groq API (LLaMA-3.1-8B-Instant), automating voice-driven workflows and reducing manual interaction by 40%.
- Implemented end-to-end supervised machine learning solutions for heart disease prediction, stock price forecasting, Bitcoin price prediction, and fake news detection using classification and regression algorithms.
- Executed complete ML pipelines covering data preprocessing, EDA, feature engineering, model selection, cross-validation, hyperparameter tuning, and deployment, reducing model development cycle time by 25%.
- Deployed 5+ machine learning applications using Streamlit dashboards for real-time inference and business automation.

Data Science Intern *Innodatatics*

May 2024 – Jun 2024

Project: Text Emotion Analysis using Large Language Models (LLMs)

- Built an LLM-based pipeline for multi-label emotion classification across 16 emotion categories, with integrated grammar scoring and filler-word detection, achieving BLEU score of 45.12 and ROUGE-L score of 67.35.
- Developed a full-stack web application using Streamlit (frontend) and PostgreSQL (backend), improving analyst workflow efficiency by 20% and reducing manual review workload by 50%.

Project: Bird Detection and Tracking using YOLOv8x

- Trained a YOLOv8x object detection model on 4,981 labeled images with age-based bird classification (juvenile vs. adult), achieving 0.96 precision, recall, and F1-score, and 0.95 mAP@0.5 on the test set.
- Built a real-time object tracking pipeline using OpenCV and deployed the system via Flask, improving detection throughput by 20% and reducing manual labeling workload by 15%.

KEY PROJECTS

ANAKIN – Real-Time Multimodal AI Assistant

2025

- Architected a voice-activated AI assistant integrating Speech-to-Text, Text-to-Speech, computer vision (OpenCV), and LLaMA-3.1-8B-Instant (via Groq API) for hands-free, real-time task automation; reduced task completion time by 40%.

Pneumonia Detection – Medical Image Classification

2025

- Developed a CNN binary classifier on NIH chest X-ray data achieving 92–95% diagnostic accuracy; applied transfer learning and class-imbalance correction to improve clinical sensitivity.

Walmart Retail Demand Forecasting

2025

- Compared ARIMA, SARIMA, and LSTM models across 45 store locations; LSTM outperformed classical baselines by 5% MAPE reduction, enabling more accurate inventory and supply chain planning.

EDUCATION

Bachelor of Engineering – Mechanical Engineering *New Horizon College of Engineering, VTU*

Aug 2014 – Jun 2021

CERTIFICATIONS

Data Science Certification *NASSCOM*

Oct 2023